Problem 1: Let $V = \mathbb{R}^3$, and let

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

(a) Prove that S is linearly independent.

(b) Extend S to form a basis for $\mathbb{R}^3$.

Solution

(a) Let $a \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \mathbf{0}$. Then:

$$a + 2b = 0, \quad b = 0, \quad a + 3b = 0 \Rightarrow a = b = 0.$$

$\Rightarrow S$ is linearly independent.

(b) Let $\mathbf{w} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and form

$$A = \begin{pmatrix} 1 & 2 & x \\ 0 & 1 & y \\ 1 & 3 & z \end{pmatrix}, \quad \det(A) = 1(z - 3y) - 2(-y) + x(-1) = -x - y + z.$$

Choose $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$: $\det(A) = -0 - 0 + 1 = 1 \neq 0 \Rightarrow$ the set is a basis.

(a) Independent; (b) Basis: $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Problem 2: Let $V = \mathbb{R}^3$, and let

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} \right\}.$$

(a) Prove that S is a linearly independent set.

(b) Extend S to form a basis for $\mathbb{R}^3$.

Solution

(a) Let $a \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + b \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} = \mathbf{0}$. Then:

$$a + 3b = 0, \quad 2a + 5b = 0, \quad b = 0 \Rightarrow a = 0.$$

$\Rightarrow S$ is linearly independent.

(b) Let $\mathbf{w} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and form

$$A = \begin{pmatrix} 1 & 3 & x \\ 2 & 5 & y \\ 0 & 1 & z \end{pmatrix}, \quad \det(A) = 1(5z - y) - 3(2z - 0) + x(2 - 0) = 2x - y - z.$$

Choose $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$: $\det(A) = -0 - 0 + 1 = 1 \neq 0 \Rightarrow$ the set is a basis.

(a) Independent; (b) Basis: $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Problem 3: Let $V = \mathbb{R}^3$, and let

$$S = \left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} \right\}.$$

(a) Prove that S is a linearly independent set.

(b) Extend S to form a basis for $\mathbb{R}^3$.

Solution

(a) Let $a \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + b \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} = \mathbf{0}$. Then:

$$2a - b = 0, \quad a + 3b = 0, \quad -a + 4b = 0 \Rightarrow a = b = 0.$$

$\Rightarrow S$ is linearly independent.

(b) Let $\mathbf{w} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and form

$$A = \begin{pmatrix} 2 & -1 & x \\ 1 & 3 & y \\ -1 & 4 & z \end{pmatrix}, \quad \det(A) = 2(3z-4y)-(-1)(z+y)+x(14-(-3)) = 7x-7y+7z.$$

Choose $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$: $\det(A) = 7 \neq 0 \Rightarrow$ the set is a basis.

(a) Independent; (b) Basis: $\left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$